

KARTIK GOYAL

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PROFESSIONAL SUMMARY

Driven Computer Science Engineering student (4th Semester) at BPIT with a 7.7 CGPA and a strong foundation in full-stack development and systems architecture. Proficient in the **MERN Stack**, **Relational Database Design (3NF)**, and high-performance **C++ programming**. Demonstrated expertise in building scalable, maintainable web applications and low-level hardware-software integration. Committed to applying algorithmic problem-solving and rigorous engineering principles to develop robust enterprise solutions.

EDUCATION

Bhagwan Parshuram Institute of Technology (BPIT), GGSIPU Delhi, India
Bachelor of Technology (B.Tech) in Computer Science and Engineering 2024 – 2028

- Current CGPA: 7.7/10
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (Java), Database Management Systems (Oracle SQL), Java Programming (CIC-212), Operating Systems, Discrete Mathematics, Computer Networks.

TECHNICAL SKILLS

Languages: C++ (Expertise in DSA), Java (Advanced Multithreading/AWT), JavaScript (ES6+), SQL, PL/SQL.

Web Development: MongoDB, Express.js, React.js, Node.js (MERN Stack), HTML5, CSS3.

Tools & Databases: Oracle RDBMS, Git/GitHub, VS Code, GitHub Codespaces, ESP32 Microcontroller.

Core Concepts: 3NF Normalization, RESTful APIs, System Design, ER Diagramming, Hardware-Software Integration.

TECHNICAL PROJECTS

University Database Management System (Oracle SQL, PL/SQL) Jan – Mar 2026

- **Architected** a comprehensive relational database for university records, designing normalized schemas up to 3NF to eliminate data redundancy and optimize storage.
- **Developed** complex PL/SQL triggers and stored procedures to automate validation and enforce strict referential integrity across multiple entities.
- **Constructed** ER diagrams to map complex relationships between Students, Faculty, and Departments, ensuring high-performance query execution.

ESP32 Electronic Voting Machine (Embedded C++, OLED) Nov – Dec 2025

- **Engineered** secure firmware utilizing interrupt-driven real-time logic to ensure instantaneous response and system stability.
- **Integrated** OLED visual feedback interfaces and implemented persistent logic to mitigate duplicate voting and enhance security protocols.
- **Streamlined** hardware-software communication protocols to ensure accurate data logging and power efficiency in a portable environment.

Attendify - Attendance Tracking System (MERN Stack) Aug – Oct 2025

- **Orchestrated** the backend architecture and database schema, focusing on scalable data flows for a collaborative management environment.
- **Facilitated** seamless team workflows and versioning by utilizing GitHub-based collaborative development and Codespaces.
- **Optimized** RESTful API endpoints to handle concurrent user requests with minimal latency and high reliability.

- **Spearheaded** the development of a Color Scheme Switcher using advanced DOM manipulation for dynamic, real-time UI updates based on user input.
- **Constructed** a logic-based Number Guessing Game focusing on efficient state management and responsive design across various screen dimensions.

ACHIEVEMENTS & CERTIFICATIONS

- **DSA Milestone:** Solved 50+ complex Data Structures and Algorithms problems in C++ (Love Babbar series), mastering linked lists, trees, and recursion.
- **Technical Training:** Completed intensive coursework in MERN Stack Development and Java Programming (CIC-212) under specialized faculty mentorship.
- **Academic Consistency:** Maintained a stable 7.7 CGPA while concurrently managing core engineering labs and independent technical research projects.